

Article

Spatial–Temporal Evolution Characteristics and Influencing Factors for the Coupling Coordinated Development of Transport Logistics and Technology

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Abstract: In recent years, China's transport logistics industry has experienced rapid development, driven by the technological advancements. But the coupling mechanism between transport logistics and technology is currently unclear, and there are likely regional differences. This study uses the entropy weight method, coupling coordination models and 20-year provincial panel data to measure the coupling coordinated development level of transport logistics and technology across 31 Chinese provinces (districts, cities). The spatial–temporal distribution, dynamic evolution, and regional differences in the coupling coordination development were analysed using kernel density estimation and the Moran index. Through the application of the Spatial Durbin Model (SDM), the mechanisms and spatial effects of selected influencing factors on the development levels are revealed. The results of this study revealed the following findings. (1) The levels of development in transport logistics and technology have consistently shown a positive upward trend with regional disparities. (2) Most provinces demonstrated a positive upward trend in the coupling coordinated development with a multi-polarised state. The overall level of coupling coordination is decreasing from east to west. In 2022, the difference between the highest and lowest coupling coordination degree between provinces is 0.78. (3) The national economy, industrial structure, urbanisation level, and consumption intensity have positive impacts on the coupling coordinated development of local regions. The findings of this study, which reveal positive trends and significant regional disparities, underscore the importance of formulating strategic plans tailored to local conditions to promote the coupled development of transport logistics and technology.

Keywords: coupling coordinated development; transport logistic; technology; spatial–temporal evolution; influencing factors



Academic Editors: Giada La Scalia and Lijun Ma

Received: 16 October 2024

Revised: 7 January 2025

Accepted: 6 February 2025

Published: 8 February 2025

Citation: Song, Q.; Ji, S.; Deng, H. Spatial–Temporal Evolution Characteristics and Influencing Factors for the Coupling Coordinated Development of Transport Logistics and Technology. *Sustainability* **2025**, *17*, 1389. <https://doi.org/10.3390/su17041389>

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1. Introduction

The transport logistics industry represents a crucial nexus between production and consumption, a composite industry that integrates transportation, warehousing, distribution, delivery, and information services [1]. Transport logistics exerts a fundamental, pioneering, and strategic influence on the development of the modern logistics system. The development of transport logistics contributes to extending industrial chains, enhancing China's position in the global value chain, and supporting the construction of a modernised economic system [2]. As per statistics, in China, the gross logistics volume of the entire society was 352.4 trillion yuan in 2023, up 5.2% year-on-year, and the total revenue of the logistics industry was 13.2 trillion yuan, up 3.9%. The annual express delivery volume

reached 132 billion, maintaining its global leadership for ten consecutive years, while the annual express logistics revenue grew by 14.3%. The ratio of total social logistics cost to GDP is 14.4%, representing a 0.3% decrease compared to 2022. While this reflects a continuous downward trend throughout the year, further efforts are needed to reduce costs in transport logistics. Currently, the logistics industry exhibits a structural imbalance, with performance stronger in the eastern and urban regions, while the western and rural areas exhibit relatively weaker performance. Overall, the transport logistics in China has achieved rapid development in recent years, but there are also problems of high production costs and unbalanced regional development [3,4]. Therefore, there are still more challenges in achieving quality, efficiency, and cost reduction while maintaining a medium-high growth rate of transport logistics.

For all industries, technology is a necessary component for achieving development goals. Therefore, the Chinese government has published numerous documents aimed at supporting the deep integration and innovative development of transport logistics and technology, thereby promoting the transport logistics industry to reduce costs, increase efficiency, and achieve stable and healthy development [5]. As the backbone of the global economy, efficient transport logistics systems are becoming increasingly integrated with technological innovations, driving operational efficiency and enabling new business models. As the demand for logistics increases, the application of technologies, such as big data, the Internet of Things, and 5G, in transport logistics has accelerated. The increasingly complex needs of logistics have also driven technological innovation, leading to the development and popularisation of new technologies and equipment, such as network freight, intelligent warehousing, automated sorting systems, and express electronic waybill. The development of transport logistics and technology is mutually reinforcing, as technological advancements drive logistics efficiency, while evolving logistics demands foster further technological innovation; the interaction and synergistic development between the two forms a virtuous circle of mutual promotion. Therefore, accelerating technological innovation and upgrading in transport logistics has become a necessity in the context of the new round of scientific and technological revolution. This transformation aims to create a new model of intelligent transport logistics characterised by advanced technologies and enhanced innovation capacity.

However, the current development of China's transport logistics is uncoordinated, with levels progressively declining from the eastern to the central-western regions. This regional imbalance is largely driven by disparities in technological adoption and innovation, which significantly influence logistics efficiency and sustainable development [1,6]. This underscores the fact that transport logistics and technology are intrinsically linked within the context of sustainable development. However, there is still a pressing need for an in-depth study of their coupling and coordination, and how they can work together to effectively address regional disparities. The coupling coordinated development of transport logistics and technology is crucial for digitally empowering logistics to enhance efficiency and reduce costs. The term "coupling" refers to the interaction between two or more complex subsystems, while "coordination" emphasises harmonious interaction among these subsystems [7]. Notably, the differences in technological advancement and logistics capabilities across spatial and temporal dimensions are significant, reflecting the complexity and dynamism of their coupling and coordination. Therefore, analysing their coupling and coordination from a spatial-temporal perspective can help identify regional disparities and points of coupling imbalance. This provides decision-making support for governments and enterprises, enabling the formulation of effective policies and strategies to avoid resource waste and promote balanced regional development. Moreover, there are many differences within areas in terms of economic development, energy reserves and geographic

location. Scientific decision-making for transport logistics and technology must be tailored to specific situations, as a uniform approach may disrupt the coordinated development between transport logistics and technology. Despite the current extensive research on transport logistics and technology individually, it mainly focuses on the impact of general or logistics-specific technology on transport logistics, while neglecting the dynamic interaction and spatial–temporal coupling coordinated relationship between them. This oversight underscores the urgent need for further comprehensive theoretical and empirical research on the coupled development of technology and transport logistics.

Therefore, this study aims to construct a comprehensive indicator for measuring the quality development of transport logistics and technology, as well as a model for their coupling coordinated development. The objective is to address these gaps by systematically clarifying the spatial–temporal distribution patterns, development trends, and driving mechanism behind their coupling coordinated development. By doing so, this study will reveal how transport logistics and technology can collaborate to reduce regional disparities and achieve sustainable development goals. This not only facilitates the in-depth integration and implementation of high-quality development strategies but advances the transformation and upgrading of China’s transport logistics, contributing to the formation of a new development pattern.

This paper is organised as follows: Section 2 reviews the recent literature; Section 3 describes the theoretical approach used in this study; Section 4 analyses the spatial–temporal evolution characteristics of the coupling coordinated development between transport logistics and technology; Section 5 examines the factors influencing the coordinated development of transport logistics and technology; Section 6 concludes the study.

2. Literature Review

In research on the high-quality development of transport logistics, many scholars have explored the level of development, spatial correlations, and influencing factors from various perspectives. There are multiple methods to measure the level of development in transport logistics, including data envelopment analysis (DEA) [8], stochastic frontier analysis (SFA) [9], fuzzy comprehensive evaluation (FCE), and the entropy weight method (EWM). Although these methods are widely used, they have certain limitations and often fail to fully capture the complexities of logistics systems, particularly regarding dynamics and soft factors. Therefore, a more integrated approach combining multiple methods is essential. The entropy weight method, Moran index, and spatial econometrics have been used to identify the spatial agglomeration and dynamic evolution characteristics of logistics development, revealing significant regional disparities [6]. To analyse the impact of each factor, a geographical and temporal weighted regression model was employed [1]. Additionally, scholars have added the background of a low-carbon [9,10] and demand economy [11] to explore the spatial–temporal evolution and influencing factors in transport logistics. These studies similarly conclude that there is spatial heterogeneity and obvious stage characteristics in the development level of the logistics industry. However, current research has focused mainly on a quantitative analysis of logistics scale and infrastructure, while insufficient attention has been paid to soft factors, such as technological and cultural aspects. Additionally, there is a lack of focus on the development of transport logistics within the context of a high-quality development framework [12].

Innovative technologies in transport logistics have significantly improved the transparency, efficiency, and security of logistics operations, driving the modernisation and global competitiveness of the industry [13]. Many studies show that innovative capacity and technology directly enhance both the overall logistics system and its individual components. This underlines the strategic importance of enhancing scientific and technological

standards in the logistics industry [14]. However, the widespread use of digital logistics technology is hampered by a high initial investment and a lack of technical expertise [15]. Many scholars have deeply explored the application and role of technology in transport logistics from different perspectives and, thus, promoted research on the impact of technology in the field of transport logistics. Nan et al. evaluated the green innovation capacity of the logistics industry in 31 provinces in China using a spatial econometric model [16]. The results show a strong positive geographical correlation in green innovation capacity, with uneven development across regions significantly influenced by economic conditions, government support, industrial structure, and innovative talent. Cheng et al. used an econometric model to analyse the impact of key factors, such as technology innovation, on the development level of a green logistics industry in China [17]. The study shows that technology innovation significantly improves logistics efficiency and environmental performance, especially in areas with high-tech investment, but the impact gradually weakened after a certain level. However, existing research is still inadequate in exploring the coupled development of transport logistics and technology, and further research into the synergies and their necessity for the future development of the industry is fundamental.

The coupling of the coordinated development between logistics and multiple fields has consistently attracted scholarly attention. The coupling of the coordinated efficiency of manufacturing and logistics in the Yellow River Basin was revealed based on the super-efficient SBM-undesirable model, and the spatial interaction effects of the two were analysed using Moran's I and the SDM [18]. The level of the coordinated development of modern logistics and tourism was measured using the entropy weight method and the coupling coordination model, and drawing the spatial evolution map of cold-hot spots to reveal the spatial effect of their coupled development [19]. The same approach was employed to explore the coordinated development of port logistics and international trade in six coastal provinces of China [20] and of port logistics and hinterland economy in Shenzhen [21]. In terms of transport logistics and technology, most scholars are relatively consistent in their views and agree that there is an interactive relationship between them. Studies on the dynamic response mechanism between green logistics, technological innovation, and the green economy have shown that green logistics is a catalyst for technological progress with a lag effect [22]. Analysis in the Yangtze River Economic Belt has confirmed that innovation and talent significantly boost the coupled development of science, technology finance, and the logistics industry [7]. However, much of the current research focuses on the coupling of logistics with other single fields and does not sufficiently consider the coupled coordination between logistics and technology within a broader system. Therefore, future research should focus on the deeper coupling mechanisms between logistics and technology, exploring their role in driving innovation and sustainable development within the industry.

Although the coordinated development of transport logistics and technology has garnered increasing attention, significant research gaps still exist. Firstly, previous studies often rely on a single indicator to quantify the development level of transport logistics and technology, neglecting the spatiotemporal dimension. This limitation hinders a comprehensive understanding of the geographic and temporal variations in their coupling. Secondly, the literature reviews show that domestic and foreign scholars study the high-quality development of transport logistics or the application of particular technology in the logistics industry. While some scholars have begun exploring the coupling of transport logistics with other sectors, the majority of studies focus on the perspectives of regional economies, energy, and environment. There is still a lack of research on the direction of the coupling coordinated development of transport logistics and technology. Thirdly, a key research gap is the lack of a comprehensive theoretical framework for analysing this

coordination. Although some studies have explored the coupling, an integrated framework that combines multiple methods is needed to better understand the dynamic relationship between these two sectors.

In response to these research gaps, this paper commits to constructing a comprehensive evaluation system and theoretical framework. It aims to reveal the spatiotemporal evolution and influencing factors of the coupled and coordinated development between transport logistics and technology. This study first constructs the evaluation index system of high-quality development in three dimensions: development investment, development construction, and development outcome. Secondly, the indicator data of 31 provinces in China from 2003 to 2022 are employed to measure and evaluate the development level and coupling coordination level of transport logistics and technology. Then, further on this basis, an exploration of the spatial distribution and dynamic evolution of the coupling coordinated development will be conducted. Finally, a system of influencing indicators is constructed to explore the factors influencing the coupling coordinated development.

3. Research Design

3.1. Index System Construction for High-Quality Development

Before calculating the coupling coordinated development of transport logistics and technology, it is essential to establish a system of indicators to calculate the development levels of both fields. In selecting indicators, the characteristics of transport logistics and technology development are fully considered, adhering to the principles of scientificity, operability, systematicity, and simplicity. Construct three primary indicators from the perspective of high-quality development: development investment, development construction, and development outcome. Development investment focuses on financial investment and human capital, assessing the availability of financial resources and skilled personnel for driving sustainable growth. Development construction emphasizes the development process, highlighting the infrastructure that supports industry operations and efficiency. Development outcome tracks the tangible results, such as market share and product quality, reflecting the tangible results and effectiveness of development strategies.

The research refers to the research results of the evaluation index for high-quality development of the logistics industry [1,6,19] and science and technology [23]. Considering the representativeness, economy, and accessibility of the data, this study selects appropriate index data as the second-level indicators of the evaluation index system for high-quality development of transport logistics and technology, as shown in Table 1. The “Properties” in the table represent the mutual relationship between the secondary and primary indicators. The “Weight” reflects the proportion of each secondary indicator in relation to the overall analysis.

3.2. Research Area and Data Source

This study adopts statistical data and space vector data from 2003 to 2022. The data sources are mainly from the China Statistical Yearbook, the China Science and Technology Statistical Yearbook, the China Financial Yearbook, the China High-Tech Industry Statistical Yearbook, the China Fixed Asset Investment Statistical Yearbook, and the website of the National Bureau of Statistics. Given the availability of data and the extensive scope of transport logistics, individual indicators are represented using data from the transportation, warehouse, and postal sectors [24]. To ensure data completeness, interpolation methods are applied to supplement the few missing values based on the existing data. Since the study primarily draws on national statistical sources, the research data is largely stable and reliable.

Table 1. The evaluation index system for the high-quality development of transport logistics and technology.

System Level	First Level Indicators	Second Level Indicators	Meaning and Calculation Method	Properties	Weights
Transport logistics	Development investment 7.281	Industrial investment (X_1)	Ratio of regional investment in fixed assets in transportation, warehousing, and postal services to regional investment in fixed assets	+	3.107
		Human capital (X_2)	Ratio of the number of persons employed in the road, river, rail, and postal sectors in the region to permanent population at the end of the year in the region	+	4.174
	Development construction 32.503	Scale of road network (X_3)	Ratio of the length of total regional road, river, and rail transport routes to the area of the region	+	3.836
		Scale of service facilities (X_4)	Ratio of the number of postal outlets to the area of the region	+	18.19
		Scale of transport facilities (X_5)	Sum of the tonnage of goods vehicles operating on roads and the net deadweight of civilian motor vessels (ton)	+	10.477
	Development outcome 60.206	Scale of transport demand (X_6)	Volume of goods transported by road, river, and rail (10,000 tons)	+	4.698
		Scale of postal requirements (X_7)	Total postal operations (100 million yuan)	+	19.549
		Scale of express demand (X_8)	Total express delivery operations (10,000 pieces)	+	26.273
		production activity (X_9)	Road, river, and rail freight turnover (billion ton-km)	+	7.877
		Operational performance (X_{10})	Ratio of regional transportation, warehousing, and postal growth to regional GDP	+	1.809
Technology	Development investment 21.484	Local financial support (X_{11})	Ratio of local finance expenditure on science and technology to local finance expenditure	+	8.058
		Full-time R&D staff (X_{12})	Ratio of full-time equivalents of R&D personnel to the year-end resident population of the region	+	13.426
	Development construction 9.6	R&D expenditure intensity (X_{13})	Ratio of R&D investment to regional GDP	+	7.552
		Intensity of service for teachers and students in higher education (X_{14})	Ratio of the number of students enrolled in general higher education institutions to the total number of faculty members	+	2.048
	Development outcome 68.915	Patents (X_{15})	Domestic patent applications received	+	24.829
		Papers (X_{16})	Number of scientific and technical papers included in major foreign search tools	+	18.386
		Technical market outputs (X_{17})	Ratio of technology market turnover to regional GDP	+	23.701
		Profitability of production and operation (X_{18})	Ratio of production and operating profit of high-tech industries to regional GDP	+	1.999

Note: “+” stands for positive indicator in the table.

According to availability, this study takes 31 provinces (districts and cities, excluding Hong Kong, Macao, and Taiwan) as the study area. The eastern region includes Beijing, Tianjin, Hebei, Liaoning, Shanghai, Jiangsu, Zhejiang, Fujian, Shandong, Guangdong, and Hainan. The central region includes Shanxi, Jilin, Heilongjiang, Anhui, Jiangxi, Henan, Hubei, and Hunan. The western region includes Inner Mongolia, Guangxi, Chongqing, Sichuan, Guizhou, Yunnan, Tibet, Shaanxi, Gansu, Qinghai, Ningxia, and Xinjiang. Panel data integrates changes in the time and individual dimension to better capture dynamic changes in the data and inter-individual differences.

3.3. Research Methodologies

This study focuses on the quantitative relationship of the coupling coordinated development between transport logistics and technology. The research framework is divided into three main steps. First, the development levels of transport logistics and technology are quantified through the construction of the assessment system, entropy weighting method, and linear weighting method. Second, the coupling coordination model is used to calculate their spatial–temporal coupling coordinated development. Kernel density estimation and spatial autocorrelation analysis are further applied to analyse the spatial–temporal dynamic evolution. Third, the spatial Durbin model is applied to identify the factors influencing their coupling coordinated development. The technical route of this study is shown in Figure 1.

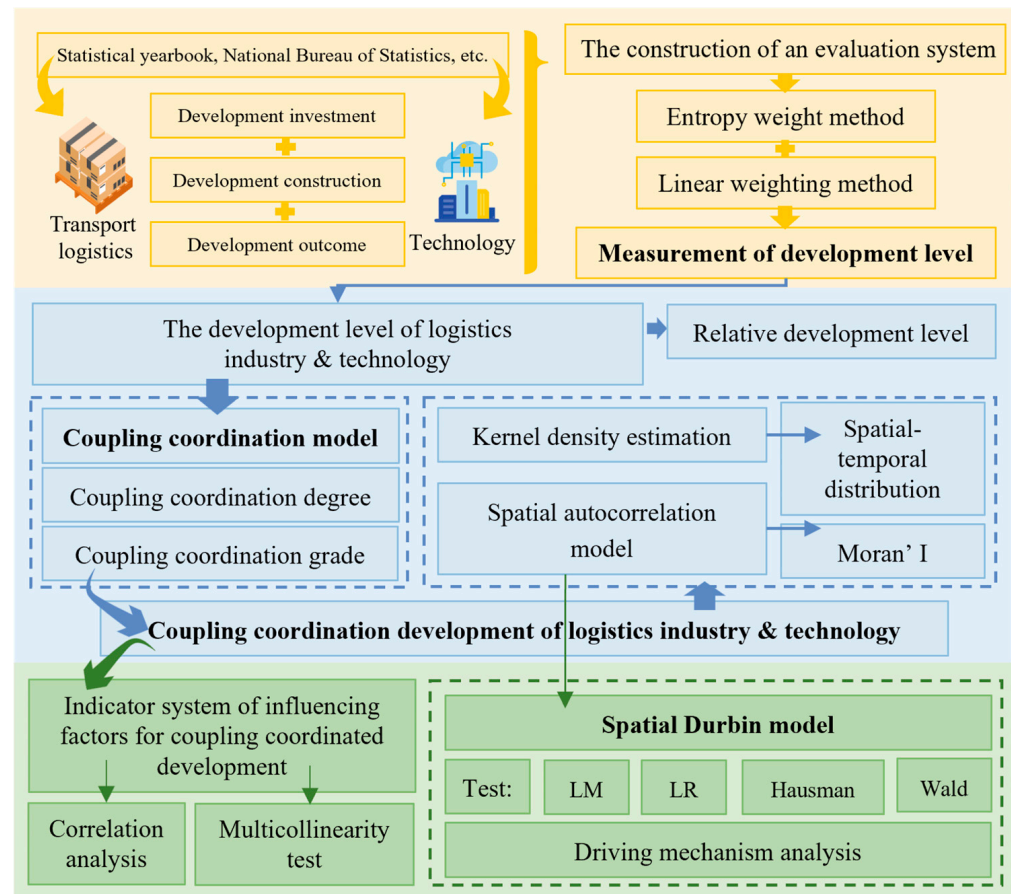


Figure 1. Technical route for quantifying and analysing the coupling coordinated development.

3.3.1. Measurement of Development Level

The evaluation indicator system utilises the entropy weight method (EWM) to calculate the weight of the individual indicator, enabling an objective assessment of the overall level of development. By analysing the inherent variability within the data, the EWM assigns higher weights to indicators with greater variability, reflecting their relative significance in the evaluation process. This method overcomes the subjective drawbacks of the hierarchical analysis method (AHP), making the calculated weights more scientific and easier to understand and accept [6].

In order to eliminate the influence of the data indicators' outline and level on the subsequent analysis, the original data of the indicators were uniformly dimensionless using the method of extreme deviation.

$$u_{ij} = \begin{cases} \frac{\max x_{ij} - x_{ij}}{\max x_{ij} - \min x_{ij}}, & x_{ij} < 0 \\ \frac{x_{ij} - \min x_{ij}}{\max x_{ij} - \min x_{ij}}, & x_{ij} > 0 \end{cases} \quad (1)$$

where x_{ij} is the j -th indicator in the i -th sample, $\min x_{ij}$ and $\max x_{ij}$ are the min-value and max-value of the j -th indicator, and u_{ij} is the normalised value.

$$p_{ij} = \frac{u_{ij}}{\sum_{i=1}^n u_{ij}}, \quad i = 1, 2, \dots, n; j = 1, 2, \dots, m \quad (2)$$

$$e_j = \frac{-\sum_{i=1}^n p_{ij} \times \ln p_{ij}}{\ln n}, \quad j = 1, 2, \dots, m \quad (3)$$

$$w_j = \frac{1 - e_j}{\sum_{j=1}^m 1 - e_j}, \quad j = 1, 2, \dots, m \quad (4)$$

$$U_i = \sum_{j=1}^m w_j \cdot u_{ij}, \quad i = 1, 2, \dots, n \quad (5)$$

In the formula, p_{ij} is the characteristic weight; e_j is the entropy value of the indicator, judging the degree of dispersion; w_j is the weight of each indicator; the comprehensive evaluation index U_i of each sample is measured using a linear weighting method. The results are shown in Table 1. The respective levels of development in transport logistics and technology have been successfully quantified.

3.3.2. The Coupling Coordination Model

Construct a coupling coordination model to calculate the coupling degree and the coupling coordination degree (CCD) for the development levels of transport logistics and technology as follows:

$$C_i = \left\{ \frac{U_1 \times U_2 \times \dots \times U_i}{[(U_1 \times U_2 \times \dots \times U_i) / i]^i} \right\}^{1/i} \quad (6)$$

$$D_i = \sqrt{T \times C_i}, \quad T = a_1 U_1 + a_2 U_2 + \dots + a_i U_i \quad (7)$$

U_i is the comprehensive evaluation index of the development of each system. T represents the comprehensive evaluation index of the coupling coordinated relationship between the systems. $a_1, a_2, \dots, a_i$ represent the coefficients to be determined, and their sum value is 1, which represents the weight of each system in this comprehensive evaluation system and can be used to identify the relative importance of the development of subsystem. In this study, $a_i = 0.5$ is taken here for both transport logistics and technology as they are equally important in social development.

In this paper, the common criteria for classifying the coupling coordination degree are used as the criteria for classifying the grades, and the 10 grades are divided into 3 levels of the coupling coordinated development.

Although the coupling coordination model can reflect the degree of the coupled development between the two systems, it cannot measure the development gap between them. Therefore, it is also necessary to introduce a relative development degree model to reflect the degree of advanced or delayed development between the two.

$$P = U_1 / U_2 \quad (8)$$

P is the relative development degree. When $P > 1.2$, the logistics industry is ahead of technology; when $0.8 < P \leq 1.2$, the transport logistics and technology are developing in tandem; when $P \leq 0.8$, the logistics industry is lagging behind [25].

3.3.3. Kernel Density Estimation (KDE)

KDE is a model for detecting data distributions in economics, and it has been widely used in spatial disequilibrium analysis [1]. This study uses KDE to judge the characteristics of the coupled development of transport logistics and technology, including the number of wave peaks, the displacement of the main peak, the changing trend of main peaks, the extension, and the polarisation trend. This method facilitates the visualisation of the CCD distribution pattern, offering deeper insights into the data's characteristics, trends, and the overall progression of coupling development levels.

$$f(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x_i - \bar{x}}{h}\right) \quad (9)$$

$$K(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) \quad (10)$$

$f(x)$ is the kernel density estimation function. Where n is the number of observations of the sample data; h denotes the bandwidth, which is closely related to the smoothness and accuracy of the kernel density curve; $K(\cdot)$ is the kernel function, which mainly includes the exponential, Gaussian, and triangle; x_i is the independent and identically distributed observations; $\bar{x}$ is the sample mean.

3.3.4. Spatial Autocorrelation Model

Kernel density analysis mainly highlights the overall distribution trend of the CCD, whereas spatial autocorrelation analysis can further reveal regional characteristics and interrelationships from a geospatial perspective. The coupled development of transport logistics and technology is analysed in space and time, with two spatial correlation test analyses, global and local. This method offers a basis for studying the imbalance of regional development.

$$W = \begin{bmatrix} w_{11} & \cdots & w_{1n} \\ \vdots & \ddots & \vdots \\ w_{n1} & \cdots & w_{nn} \end{bmatrix}, w_{ij} = \begin{cases} 1, i, j \text{ are adjacent} \\ 0, i, j \text{ are not adjacent} \end{cases} \quad (11)$$

A geographically adjacent spatial weight matrix is used, with w_{ij} denoting the distance of spatial weight between provinces, for which provinces i and j are 1 when they are adjacent and 0 when they are not.

$$Global Moran's I = \frac{n \sum_{i=1}^n \sum_{j=1}^n w_{ij} (y_i - \bar{y})(y_j - \bar{y})}{\left(\sum_{i=1}^n \sum_{j=1}^n w_{ij} \right) \sum_{i=1}^n (y_i - \bar{y})^2} \quad (12)$$

$$Local Moran's I = \frac{n(y_i - \bar{y}) \sum_{j=1}^n w_{ij} (y_j - \bar{y})}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (13)$$

where n is the number of provinces, and $y_i, y_j, \bar{y}$ denote the coupling coordinated degree of two systems, respectively, and the mean value.

3.3.5. Spatial Durbin Model (SDM)

Based on the spatial autocorrelation analysis, the SDM is employed to identify the key factors influencing the coupling coordinated development of transport logistics and

technology. This method provides a powerful tool for understanding regional development disparities and formulating policies to promote regional coordinated development. The SDM incorporates both exogenous and endogenous spatial dependence, considering the spatial correlation between the explained and explanatory variables.

$$Y = \gamma + \rho WY + \beta X + \varphi WX + \varepsilon \quad (14)$$

where Y is the dependent variable, i.e., the explained variable; X is the independent variable, i.e., the explanatory variable; γ is a constant term; ρ is the spatial autoregressive coefficient, which indicates the effect of the dependent variable in the neighbouring region on the local dependent variable; β and φ are both a spatial correlation coefficient, which indicates the effect of the independent variable in the neighbouring region on the local dependent variable; W is the spatial weight matrix, and ε is a random error term.

4. The Spatial Effect of the Coupling Coordinated Development

4.1. Development Level of Transport Logistics and Technology

A comprehensive evaluation index system for the development of transport logistics and technology is constructed from three dimensions (development investment, development construction, development outcome). Take the 31 provinces as the research areas for in-depth analysis. Utilise the method described in Section 3.3.1 to compute the comprehensive evaluation indexes for the high-quality development of transport logistics and technology. Figure 2 illustrates the development levels of the 31 provinces from 2003 to 2022, revealing similar developmental trends across the regions.

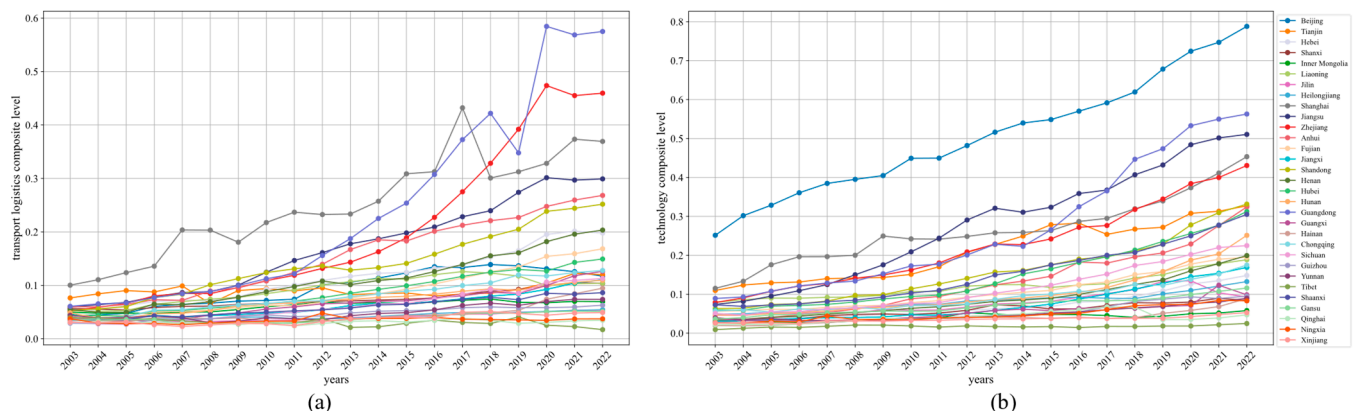


Figure 2. The development level of logistics industry and technology. (a) Transport logistics composite level of the 31 provinces from 2003 to 2022. (b) Technology composite level of the 31 provinces from 2003 to 2022.

In terms of the trend of change, the development level of transport logistics and technology have maintained a positive upward trend over the study period, with no significant upward trend of the development level in a few regions. Nationally, the development level of transport logistics rises from 0.047 in 2003 to 0.15 in 2022, with an average annual growth rate of 6.30%. The evolutionary process of transport logistics can be roughly divided into three stages, the first phase is the slow growth stage (2003–2008), with an average annual growth rate of 4.65%; the second phase is the adjustment development stage (2009–2016), with an average annual growth rate of 7.57%; and the third phase is the steady growth stage (2017–2022), with an average annual growth rate of 4.92%. The development level of technology rises from 0.056 in 2003 to 0.225 in 2022, with an average annual growth rate of 7.59%. The average annual growth rates of technology in the three corresponding phases were 7.39%, 7.64%, and 8.45%, respectively, showing a trend of increasing growth rates.

At the provincial levels, there are regional disparities in the development of transport logistics and technology. The western provinces significantly lag behind the central and eastern regions, and the gap continues to widen. The development of transport logistics and technology in the four northwestern provinces (Gansu, Ningxia, Qinghai, Xinjiang), and even the western region remain at a lower level. By contrast, the eastern coastal provinces, such as Guangdong, Zhejiang, Shanghai, and Jiangsu, exhibit higher levels of development, with Beijing consistently leading in technology. The result is consistent with the current situation for transport logistics development and technology development in China. Although China's transport logistics have developed to a great extent, the development level in the eastern region is higher than in the central and western regions, showing an evident regional imbalance. The eastern coastal provinces, with their favourable geography, support the strategic planning and robust construction of transport infrastructure. This facilitates efficient industry and resource consolidation and enhances inter-regional economic communication. By contrast, the rugged terrain of the western inland provinces significantly hampers infrastructure development, limiting the centralisation of industries and resources. Such geographical constraints hinder economic interactions and exacerbate talent migration, thereby curtailing the development of transport logistics and technology. Based on this, the evaluation results obtained from the indicator system in this paper reflect the development of China's transport logistics and technology, showing the reasonableness and applicability of the evaluation indicator system.

Figure 3 depicts the relative development degree of transport logistics and technology in the 31 provinces from 2003 to 2022, and the corresponding box-and-line diagrams are shown on the right. Relative development levels above 1.2 signify advanced logistics, those below 0.8 indicate lagging, and levels between 0.8 and 1.2 denote synchronisation. The figure illustrates that the relative development levels of many provinces are below 1.2, with an increasing number of provinces falling below 0.8. During the study period, the development level of the transport logistics in most provinces significantly lagged behind that of technology, with the number of such provinces increasing. Consequently, transport logistics and technology did not achieve the desired equivalence. The number of provinces with advanced transport logistics decreased from 10 in 2003 to 4 in 2022, and the lagging state grew from 10 to 18 regions. Among them, Inner Mongolia and Hebei have been in an advanced state, while Beijing, Shanghai, Tianjin, Hubei, Heilongjiang, and Sichuan have been lagging. Tibet entered a synchronised state from an advanced state in 2021 and is lagging in 2022. Jiangxi and Hunan shifted from a synchronised to a lagging state in 2016 and have remained in the lagging state since then.

The reason for this may be that Beijing, Shanghai, and Tianjin belong to urban units. Given the small jurisdiction area, the high population density, the large GDP, and a focus on emerging technological industries, transport logistics tends to lag behind technology. While, for other provinces belonging to regional units, differences in economic status and logistics needs lead to the complex development of transport logistics and technology. The economic structure of Hebei is relatively traditional, dominated by heavy industry, manufacturing, and agriculture, with a weak reliance on high technology. Particularly in the Beijing–Tianjin–Hebei region, Hebei suffers from serious brain drain, resulting in a lagging situation of technology lagging far behind transport logistics. Tibet's high altitude and complex terrain limit the introduction of high-tech equipment and professionals, bringing major challenges to technology development. And its economy depends mainly on agriculture, animal husbandry, and tourism, with a low dependence on technology, so technology lags in transport logistics.

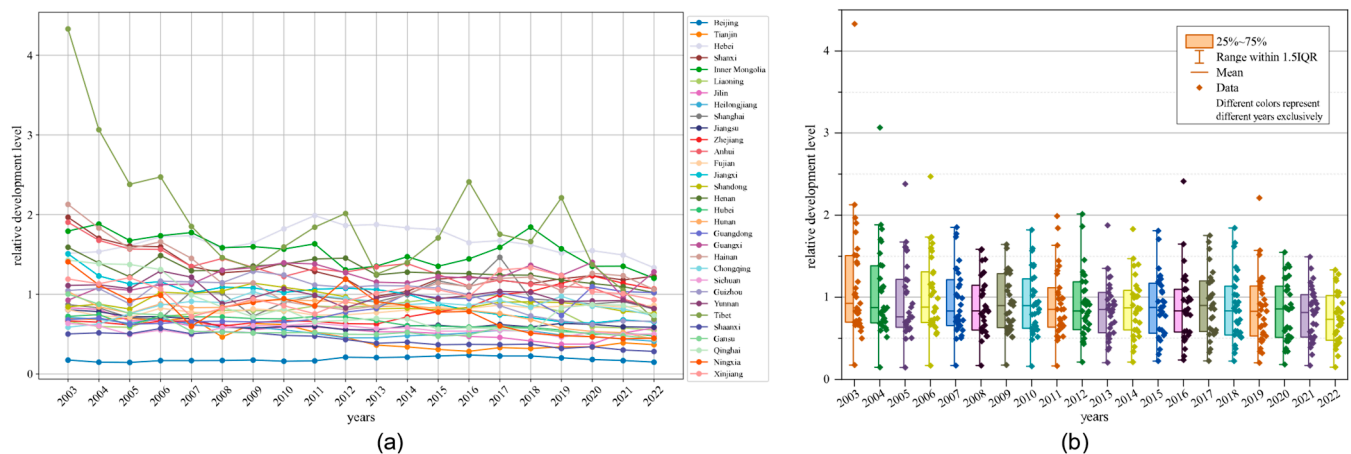


Figure 3. The relative development level of logistics industry and technology. (a) Line graph of the relative development of the 31 provinces from 2003 to 2022. (b) The corresponding box-and-line diagram for (a).

4.2. Coupling Coordinated Development of Transport Logistics and Technology

4.2.1. Spatial–Temporal Evolution Characteristics

Use the method described in Section 3.3.2 to compute the CCD of transport logistics and technology across the 31 provinces.

Figure 4 depicts the CCD of transport logistics and technology in the 31 provinces from 2003 to 2022, and the corresponding box-and-line diagrams are shown on the right. From the trend of change, the CCD of transport logistics and technology in most provinces have maintained a positive upward growth trend during the study period. The average CCD increased nationally by 3.49%, from 0.243 in 2003 to 0.466 in 2022. In 2003, the number of primary and moderate coupling development level provinces was 27 and 4, respectively. In 2022, the number of moderate coupling provinces increased to 22, and advanced coupling provinces increased to 4. However, certain regions displayed stable or even declining trends. In terms of the three stages of the evolutionary process of transport logistics development, the average annual growth rates are 2.95% (2003–2008), 3.86% (2009–2016), and 3.26% (2017–2022).

Based on the results of the CCD, from 2003 to 2008, the CCD was predominantly ranged within the interval of (0 ~ 0.3) with a more concentrated distribution, at a primary coupling level. This suggests that the correlation between the three dimensions of transport logistics and technology systems was not close. From 2009 to 2016, the correlation between these two systems improved slightly, but the distribution was still basically stable at [0.2 ~ 0.4]. This phase was characterised by mild disorder and on the verge of disorder, with fewer provinces reaching the advanced coupling level. From 2016 to 2022, it is basically at [0.3 ~ 0.5], at the moderate coupling level.

Referring to Table 2, Figure 5 shows the grades of the coupling development of transport logistics and technology in the 31 provinces from 2003 to 2022.

Based on the analysis of Figures 4 and 5, it is evident that the coupling coordinated development between transport logistics and technology exhibits a regional disparity across provinces, with a tendency to expand gradually. Currently, the coupled development across the 31 provinces predominantly remains at a moderate level. During the study period, Beijing, Tianjin, Shanghai, and Guangdong have consistently maintained a moderate to advanced level of coupled development. Tibet, Qinghai, Xinjiang, Ningxia, and Inner Mongolia have been at the primary level, with a small degree of improvement, except Tibet. In 2017, Guangdong became the first province to achieve and maintain an advanced coupled development level. The coupled development levels of Zhejiang and Jiangsu

achieved a significant improvement within 16 years, rising from the primary level to the advanced level, with the largest increase. However, the development trend is slower in Inner Mongolia, Tibet, Qinghai, Ningxia, and Xinjiang.

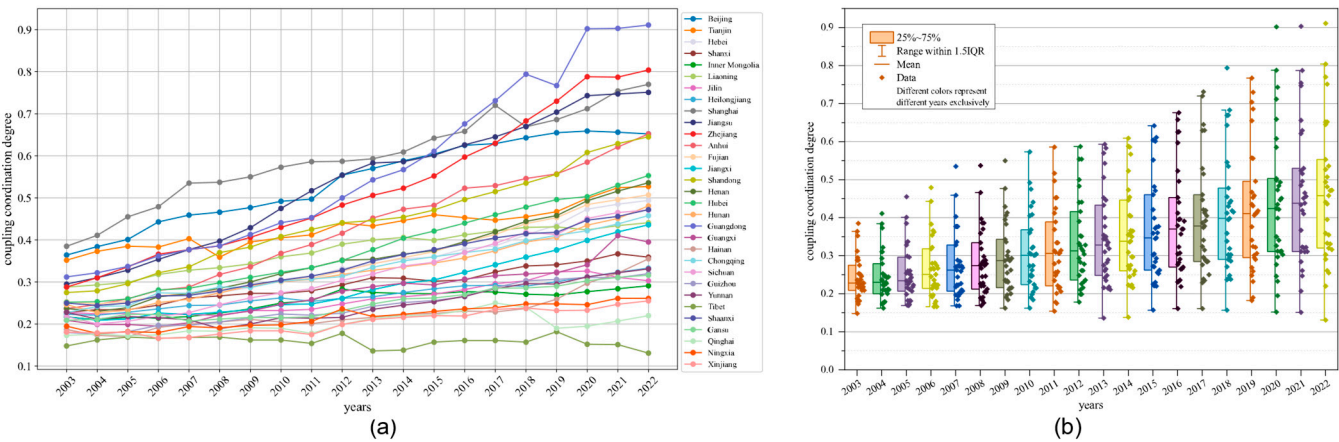


Figure 4. The coupling coordination level of transport logistics and technology. (a) Line graph of the coupling coordination level of the 31 provinces from 2003 to 2022. (b) The corresponding box-and-line diagram for (a).

Table 2. Standards for dividing the coupling coordination degree.

Coupling Coordination Degree Numerical Value	Coupling Coordination Grade	Coupling Coordination Type	Coupling Coordinated Development Level
[0.0~0.1)	1	Extreme disorder	Primary level
[0.1~0.2)	2	Severe disorder	
[0.2~0.3)	3	Moderate disorder	
[0.3~0.4)	4	Mildly disorder	Moderate level
[0.4~0.5)	5	On the verge of disorder	
[0.5~0.6)	6	Barely coordinated	
[0.6~0.7)	7	Elementary coordination	
[0.7~0.8)	8	Intermediate coordination	Advanced level
[0.8~0.9)	9	Good coordination	
[0.9~1.0]	10	Quality coordination	

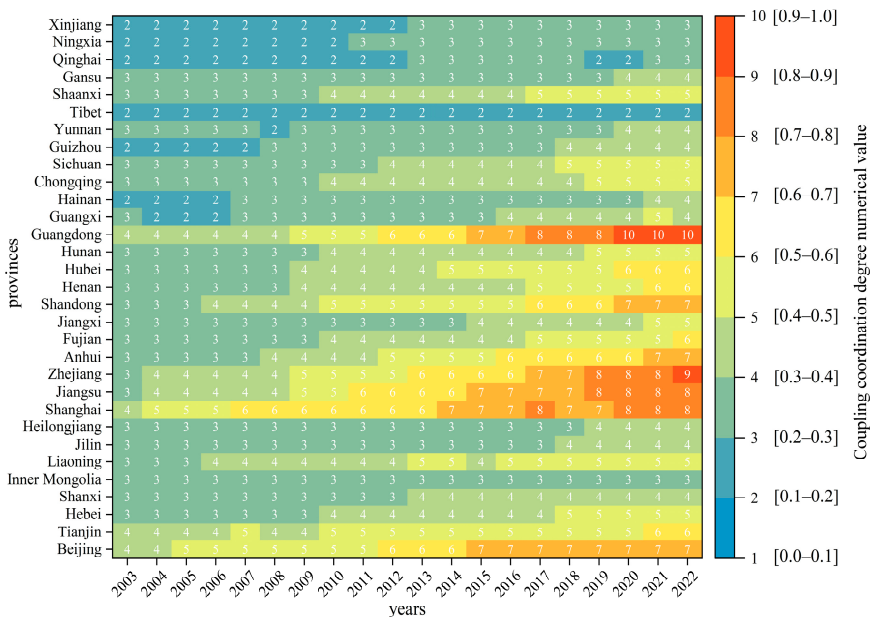


Figure 5. Grades of the coupling coordinated development of transport logistics and technology.

To further identify and compare the specific spatial distribution of the coupling coordinated development levels between transport logistics and technology in the 31 provinces from 2003 to 2022, this study combines Table 2 with ArcGIS 10.8.1 to visualise and analyse the regional characteristics and spatial heterogeneity of the coupling coordinated development levels. Figure 6 depicts the spatial distribution pattern of the coupled coordinated development between the two systems [19].

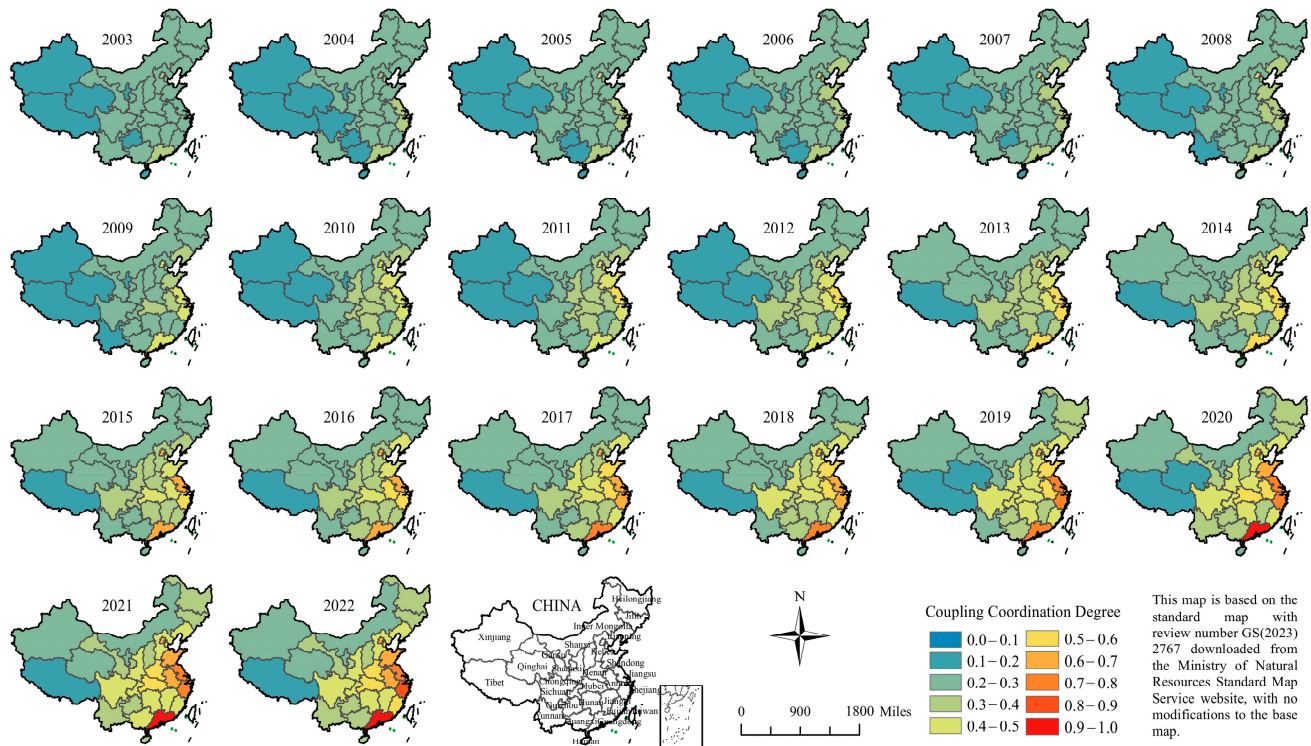


Figure 6. The spatial distribution of the coupling coordination levels of transport logistics and technology.

On a national scale, the coupling coordination level of transport logistics and technology in the 11 eastern provinces is significantly higher compared with other regions, indicating more robust and effective coupling interactions in eastern areas. Notably, Guangdong and Zhejiang not only display higher levels of coupling coordination but exhibit rapid advancements. Fujian and Hebei show lower levels of coupling coordination, while experiencing a positive trend towards improvement. By contrast, Hainan consistently demonstrates a low level of coupling coordination, with minimal fluctuations over time.

The coupling coordinated level in the eight central region is basically in the range of $[0.2 \sim 0.5]$, where the distribution is more concentrated. It suggests the coupling development level is lower, which indicates a subdued interplay between transport logistics and technology in the central region. Among these provinces, Anhui demonstrates a higher coupling level, exceeding that of Tianjin in 2013. Conversely, Heilongjiang, Jilin, and Jiangxi are at a low coupling level, and fall below the national average.

The 12 western provinces in China exhibit a low degree of coupling coordination between transport logistics and technology, indicating a weak integration in these regions. Among these provinces, Sichuan has achieved a relatively higher level of coupling, becoming the first province in the western region to reach a moderate coupling level in 2018. By contrast, provinces such as Ningxia, Xinjiang, Qinghai, and Tibet have shown slower development, with their CCD remaining within $[0.1 \sim 0.3]$.

4.2.2. Dynamic Evolution Characteristics

Based on the CCD of the coupled development of transport logistics and technology in the 31 provinces, this study selects the Gaussian kernel function with high precision to analyse the centre position, shape, and extension of the kernel density curve. It reveals the development level, distribution evolution, extension, and polarisation trend of the coupling coordinated development of the two systems more visually and intuitively. Figure 7 depicts the three-dimensional kernel density of the coupled development. In the graph, brighter colors represent higher kernel densities.

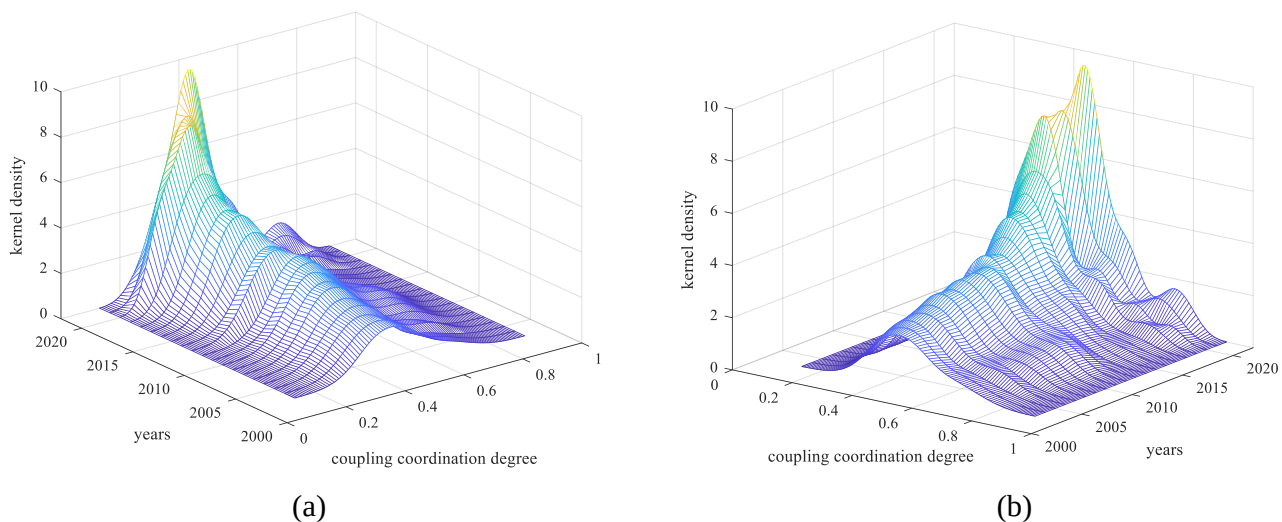


Figure 7. Dynamic evolution of the level of the coupling coordinated development. (a,b) shows two different angles of the three-dimensional kernel density.

Table 3 shows the dynamic evolution of the coupling coordination development of transport logistics and technology in the 31 provinces during the study period. First, in terms of distribution position, the centre and interval range of the kernel density curve exhibited a slight leftward shift followed by a substantial rightward shift. This indicates an upward trend in the coupling development level between the two systems. Second, in terms of peak numbers, the kernel density curve gradually evolves from a “single peak” to a “multiple peaks” form, indicating that the coupling development of transport logistics and technology is in a multi-polarised state. The increasing distance between the primary and secondary peaks indicates that, over time, the polarisation phenomenon in the coupling coordinated level between the two systems has relatively intensified. Thirdly, in terms of distribution pattern, the height of the primary peak of the kernel density curve has been showing an upward trend, and the shape of the wave crest has been gradually narrowing. This indicates that the absolute gap in the coupling development levels has been narrowing, and the degree of dispersion has been decreasing. Fourthly, in terms of distribution extensibility, the right trailing phenomenon of the kernel density curve is relatively obvious, indicating that there are large interregional differences in the level of the coupling development. Moreover, the provinces with higher levels have been maintaining rapid growth [1].

Table 3. Characteristics of the distribution of kernel density for the coupling coordinated development.

Characteristics	Dynamic Evolution
Number of peaks	single-peak—Twin-peak—multi-peak—Twin-peak
Main peak displacement	right shift—left shift—right shift
Main peak variation	Rising, narrowing
Extension	Right trailing, broadening—convergence
Polarisation characteristics	Bifurcation—Multiploidization—Bifurcation

Although the level of the coupling coordinated development of transport logistics and technology in the 31 provinces achieved different degrees of growth, the speed of the coupled development accelerated. However, due to differences in resource endowments and development environments, a number of provinces face challenges in achieving balanced development. It leads to difficulties for provinces with lower levels of coupling development to catch up with the leading provinces, ultimately widening the gap between them.

4.3. Spatial Agglomeration of the Coupling Coordination Development

4.3.1. Global Spatial Agglomeration Characteristics

To more intuitively analyse the spatial correlation and agglomeration characteristics of the coupled development levels of transport logistics and technology across the 31 provinces, the Global Moran's index is employed. This method calculates spatial autocorrelation to ascertain whether there exists any significant spatial clustering characteristics in the coupling coordinated development (see Table 4, Figure 8).

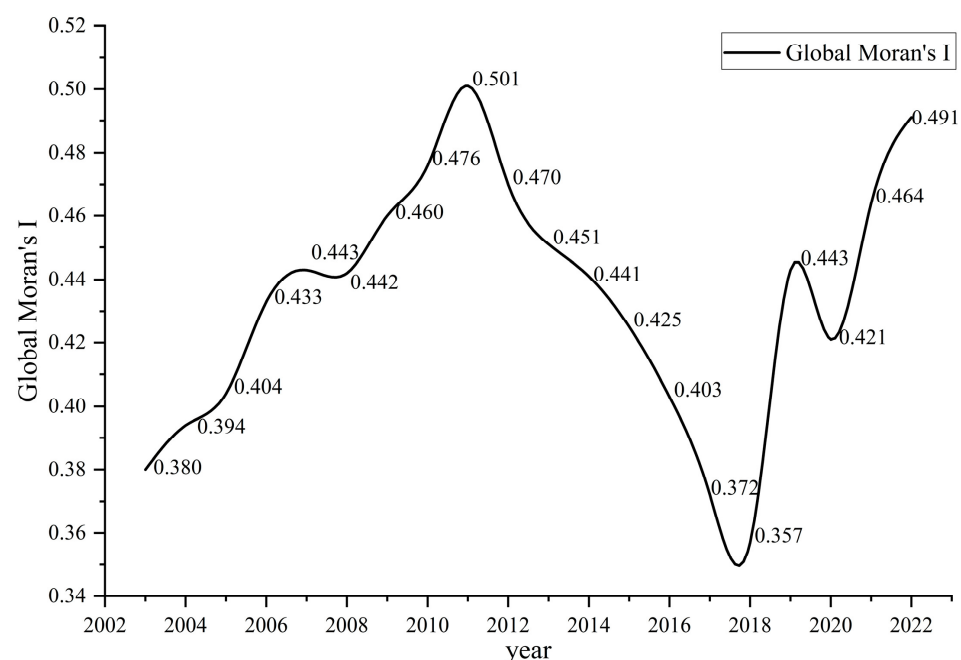
**Figure 8.** Global Moran's Index of the coupling coordinated development.

Table 4 reveals that the p -value of the Global Moran's I passes the 1% spatial significance test and fluctuates overall during the study period. It suggests that the CCD of transport logistics and technology in the 31 provinces presents a strong spatial positive correlation and obvious spatial agglomeration characteristics. Such conditions are primarily attributable to spatial spillover or expansion effects among these provinces. Therefore, it is essential to introduce spatial factors when analysing the factors influencing the coupled development of transport logistics and technology. Figure 8 shows that the Global Moran's

I increased from 0.38 to the peak value of 0.501 during 2003–2011. The Global Moran's I dropped from 0.501 to the minimum value of 0.357 during 2011–2018, which indicates a gradually weakened decrease in the spatial correlation and clustering effects of transport logistics and technology. The Global Moran's I shows a fluctuating increase during 2018–2022. The positive spatial autocorrelation of the coupling between the two systems has grown, suggesting that the coupling coordination becomes more geographically concentrated.

Table 4. Global Moran's index of the coupling coordinated development.

Years	Moran's I	z	Years	Moran's I	z
2003	0.380 ***	3.485	2013	0.451 ***	4.084
2004	0.394 ***	3.603	2014	0.441 ***	4.000
2005	0.404 ***	3.695	2015	0.425 ***	3.870
2006	0.433 ***	3.932	2016	0.403 ***	3.682
2007	0.443 ***	4.024	2017	0.372 ***	3.418
2008	0.442 ***	4.007	2018	0.357 ***	3.296
2009	0.460 ***	4.162	2019	0.443 ***	4.017
2010	0.476 ***	4.294	2020	0.421 ***	3.837
2011	0.501 ***	4.505	2021	0.464 ***	4.199
2012	0.470 ***	4.251	2022	0.491 ***	4.421

Note: *** indicates significance at the 1% level.

4.3.2. Local Spatial Agglomeration Characteristics

The Global Moran's I indicates the presence of the spatial agglomeration effect in the coupling coordinated development of transport logistics and technology in the 31 provinces. However, it is not possible to ascertain the degree of spatial correlation between each province and its surrounding provinces. This study further uses Local Moran's I scatter plot to reveal the spatial agglomeration characteristics of the coupling coordinated development. For a detailed examination of its developmental stages, the years 2003 and 2022 were selected for analysis. Figure 9 illustrates the states of spatial agglomeration in localised areas. Table 5 categorises the types of spatial agglomeration in the coupling development of transport logistics and technology across various provinces.

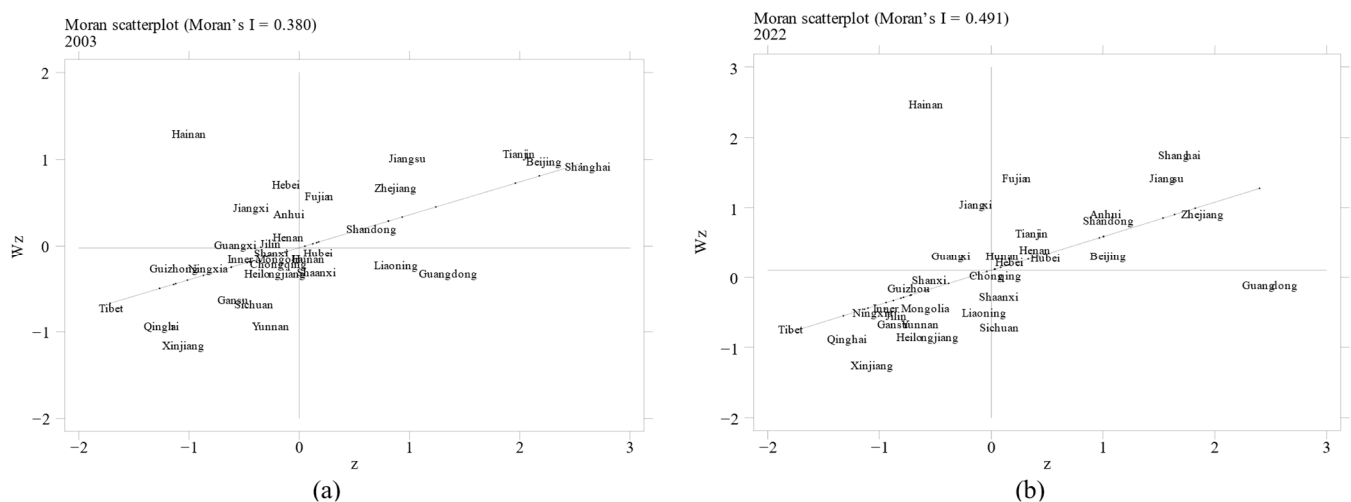


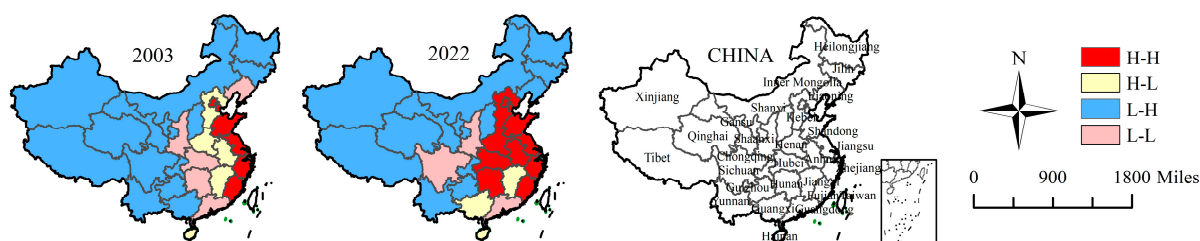
Figure 9. Local Moran's I scatter graphs of the coupling coordinated development. (a) The Local Moran's I scatter graph for 2003. (b) The Local Moran's I scatter graph for 2022.

Table 5. Spatial agglomeration types of the coupling coordinated development.

Agglomeration	2003	2022
High–High agglomeration	Beijing, Tianjin, Shanghai, Jiangsu, Zhejiang, Fujian, Shandong.	Beijing, Tianjin, Hebei, Shanghai, Jiangsu, Zhejiang, Anhui, Fujian, Shandong, Henan, Hubei, Hunan.
Low–High agglomeration	Hebei, Anhui, Jiangxi, Henan, Hainan. Shanxi, Inner Mongolia, Jilin, Heilongjiang, Guangxi, Chongqing, Sichuan, Guizhou, Yunnan, Xizang, Gansu, Qinghai, Ningxia, Xinjiang.	Jiangxi, Guangxi, Hainan. Shanxi, Inner Mongolia, Liaoning, Jilin, Heilongjiang, Chongqing, Guizhou, Yunnan, Tibet, Gansu, Qinghai, Ningxia, Xinjiang.
High–Low agglomeration	Liaoning, Hubei, Hunan, Guangdong, Shaanxi.	Guangdong, Sichuan, Shaanxi.

The spatial clustering patterns of the coupling coordinated development of transport logistics and technology in 2003 and 2022 reveal significant regional shifts. Upon a vertical comparison, the distribution pattern in 2022 closely resembles that of 2003. However, there has been a reduction in the provinces classified under the LH and HL clusters. In a cross-sectional comparison spanning from 2003 to 2022, most provinces are mainly found in either the HH cluster (first quadrant) or the LL cluster (third quadrant). This distribution suggests a positive spatial correlation in the coupling coordinated development levels of transport logistics and technology.

Figure 10 presents the results of a spatial autocorrelation analysis from a geospatial perspective. In 2003, High–High clusters were concentrated in economically advanced coastal regions. By 2022, this cluster expanded to inland areas, such as Hebei, Anhui, Henan, Hubei, and Hunan, reflecting the success of regional policies, such as the coordinated development of the Beijing–Tianjin–Hebei region, the Yangtze River Delta integration, and Central China’s rise. These regions have benefited from improved transportation infrastructure and the support of technological innovation in logistics. Meanwhile, the Low–High clusters have narrowed over time. Combined with Figure 2, this indicates that regions such as Hebei, Anhui, and Henan transitioned to High–High clusters by enhancing technological capabilities and integrating them effectively with logistics. Conversely, Low–Low clusters, concentrated in less-developed western or northeastern regions, continue to face challenges, such as limited economic scale, weak infrastructure, and insufficient technological resources. Addressing these issues requires targeted policies and substantial investment. The transitions of Hubei and Hunan to High–High clusters highlights their progress in aligning logistics with technology. However, Guangdong and Sichuan’s persistence in the High–Low clusters suggests a potential mismatch between technological progress and logistics infrastructure, necessitating strategic interventions to address these disparities.

**Figure 10.** LISA clustering maps for the coupling coordinated development.

To optimise regional development, differentiated strategies are critical. High–High regions should focus on fostering collaboration, advancing smart logistics, and promoting technological innovation. Low–High regions need to increase technological investment and

strengthen collaboration with High–High regions. Low–Low regions require focused efforts on infrastructure development, resource allocation, and leveraging regional advantages to overcome challenges. Finally, High–Low regions should prioritise enhancing logistics efficiency and better aligning it with technological advancements to achieve greater synergy.

Therefore, there is a spatial correlation in the coupling coordinated level between transport logistics technology in the 31 provinces. If this spatial effect is ignored, it may lead to biased or inconsistent parameter estimates in the impact factor study. Therefore, it is essential to examine the spatial correlation in the coupling coordinated development [6].

5. The Influencing Factors of the Coupling Coordinated Development

5.1. Index System Construction for Influencing Factors

This study refers to relevant research findings [1,26], adhering to the principles of data availability and representativeness. Following these principles, indicators are selected from a macro-level perspective to construct the influencing factor indicator system of the coupling coordinated development (Table 6).

Table 6. Indicator system of influencing factors for the coupling coordinated development.

Type	Variables	Meaning and Calculation Method	Unit
Explained variable	Coupling coordinated level (Y)	Level of coupling coordination development of transport logistics and technology.	-
Explanatory variable	National economy (x_1)	The per capita GDP.	Million yuan per person.
	Population (x_2)	Density of resident population at year-end.	Hundred persons per square kilometre.
	Industrial structure (x_3)	Ratio of value added in the tertiary industry to value added in the secondary industry.	%
	Urbanisation level (x_4)	The proportion of urban population in total population.	%
	Consumption intensity (x_5)	Consumption expenditure ratio.	%
	Foreign trade scale (x_6)	The proportion of total international trade of the business unit location in GDP.	%

This paper selects indicators from the following six aspects as influencing factors to explore how they affect the coupling coordinated development of transport logistics and technology. National economy (x_1) determines investment capacity in transportation and technology. Economically advanced regions can allocate more resources to foster innovation in transport logistics, enhancing efficiency, and integration. Economic growth also influences logistics demand, with higher activity correlating to increased reliance on efficient transport systems and advanced technologies. Population size (x_2) influences logistics demand and technology application. Densely populated areas require more robust logistics services and higher efficiency, driving the use of technology-driven solutions. Industrial structure (x_3) shapes dominant sectors and their logistical requirements, influencing the depth of technological applications. Manufacturing regions rely on efficient freight transportation and warehousing, while service-oriented economies prioritise on-demand delivery and flexible logistics, spurring innovation tailored to industry needs. Urbanisation (x_4) not only concentrates population and economic activities but increases the complexity of logistics demands. High-density urban areas require transport systems with greater operational efficiency and environmental sustainability, driving the development

of intelligent transportation systems and smart logistics. The intensity and frequency of consumption (x_5) directly reflect the demand for logistics services. High-consumption regions necessitate logistics services characterised by speed, accuracy, and convenience, which in turn drive continuous technological innovation. For example, technologies such as automated warehousing, intelligent sorting, and real-time delivery systems have emerged to meet high-frequency consumer demands. The expansion of foreign trade (x_6), driven by globalisation, places higher demands on the efficiency and reliability of cross-border logistics systems. Advanced technologies, like blockchain, IoT, and AI, enhance transparency, automation, and risk mitigation, upgrading logistics systems and boosting global collaboration. In conclusion, these six factors interact at multiple levels, forming the core driving forces behind the coupling coordinated development of transport logistics and technology. Understanding the impact of these factors on the coupling coordinated development of transport logistics and technology is essential for reducing regional disparities and promoting the sustainable development of intelligent logistics systems.

Table 7 shows the descriptive statistics of the variables. The data sources are consistent with those described in Section 3.2. According to the data in Table 7, the mean value of coupling coordination development is 0.342, with a standard deviation of 0.143, and maximum and minimum values of 0.911 and 0.131, respectively. This suggests that the overall development level remains relatively low, with significant disparities across regions. The average value of national economic development is 4.294, with a maximum of 19.00, highlighting substantial interregional disparities. Regarding population, the mean value is 4.332, with a maximum of 39.26 and a standard deviation of 0.143, indicating pronounced regional differences. By comparison, the regional disparities in urbanisation level and consumption intensity are relatively smaller.

Table 7. Descriptive statistical outcome of variables.

Variables	Obs	Mean	Medium	Std. Dev.	Min	Max
Y	620	0.342	0.303	0.143	0.131	0.911
x_1	620	4.294	3.670	3.127	0.371	19.00
x_2	620	4.332	2.705	6.509	0.022	39.26
x_3	620	1.236	1.077	0.670	0.527	5.244
x_4	620	0.539	0.533	0.153	0.202	0.913
x_5	620	0.712	0.715	0.068	0.543	0.961
x_6	620	0.302	0.140	0.355	0.008	1.711

5.2. Set-Up Tests of Spatial Durbin Model

5.2.1. Correlation Analysis of Variables

To avoid pseudo-regression problems, correlation analysis and multicollinearity tests were performed for all variables using the software Stata 16 (see Tables 8 and 9).

The correlation analysis test results show significant correlations between the explained and explanatory variables at the 1% significance level, confirming the validity of the selected explanatory variables for further empirical analysis.

Moreover, the multicollinearity test reveals that the variance inflation factor (VIF) values of all variables are below 10, with a mean VIF below 5. These findings suggest an absence of multicollinearity, thereby validating the continuation of the empirical analysis [26].

Table 8. Correlation analysis of variables.

Variables	Y	x ₁	x ₂	x ₃	x ₄	x ₅	x ₆
Y	1.000						
x ₁	0.820 ***	1.000					
x ₂	0.593 ***	0.551 ***	1.000				
x ₃	0.309 ***	0.547 ***	0.338 ***	1.000			
x ₄	0.749 ***	0.807 ***	0.610 ***	0.416 ***	1.000		
x ₅	−0.419 ***	−0.528 ***	−0.040	−0.178 ***	−0.248 ***	1.000	
x ₆	0.514 ***	0.394 ***	0.692 ***	0.304 ***	0.640 ***	0.173 ***	1.000

Note: *** indicates significance at the 1% level.

Table 9. Multicollinearity test.

Variables	x ₁	x ₄	x ₆	x ₂	x ₅	x ₃	Mean
VIF	5.72	4.60	2.90	2.35	1.88	1.51	VIF
1/VIF	0.1748	0.2175	0.3451	0.4247	0.5308	0.6632	3.16

5.2.2. Spatial Durbin Model Set-Up Tests

According to the previous analysis in Section 4.3, the spatial autocorrelation test revealed spatial agglomeration in the CCD of transport logistics and technology. In order to select an appropriate spatial econometric model, this paper employed several diagnostic tests, including a robust Lagrange multiplier (Robust-LM) test, the likelihood ratio (LR) test, the Wald test, and the Hausman test (see Table 10).

Table 10. Spatial Durbin model set-up tests.

Spatial Panels Test Model		Statistic	p-Value
LM Test	LM spatial error	14.287	0.000 ***
	Robust LM spatial error	0.268	0.605
	LM spatial lag	23.052	0.000 ***
	Robust LM spatial lag	9.033	0.003 ***
Hausman	Hausman	80.22	0.000 ***
LR Test	SDM and SAR	152.01	0.000 ***
	SDM and SEM	164.89	0.000 ***
	both and ind	48.94	0.000 ***
	both and time	690.05	0.000 ***
Wald Test	Wald SAR	170.01	0.000 ***
	Wald SEM	180.42	0.000 ***

Note: *** indicates significance at the 1% level.

The test shows that the LM-Lag and LM-Error statistics from the LM test are significant at the 10% level. This significance confirms the appropriateness of the spatial econometric model selection, suggesting that the SDM should be initially considered for analysing the impact mechanisms of coupled development. The Hausman test favours the selection of a fixed-effects model. The LR (SDM and SAR) and LR (SDM and SEM) test statistics against SAR and SEM both reject the null hypothesis at the 1% level, indicating that the SDM cannot degenerate into either an SAR or SEM. Additionally, the LR (both and ind) and LR (both and time) tests both rejected the original hypothesis at the 1% level, supporting the efficacy of a dual fixed-effects model in the SDM approach. The Wald test, significant at the 1% level, further supports the selection of the SDM as preferable. Based on this, the SDM with dual fixed-effects effects is chosen for the analysis [27].

5.3. Analysis of the Spatial Regression Results

The coefficients of the SDM do not directly reflect the spillover effects of the corresponding explanatory variables on the explained variables [28]. So, the spatial effects need to be tested in terms of total, direct, and indirect effects (see Table 11).

Table 11. Estimation and test results based on the SDM for the influencing factor.

Variables	Direct	Indirect	Total
National economy (x_1)	0.012 *** (4.93)	0.049 *** (9.68)	0.061 *** (11.16)
Population (x_2)	−0.008 *** (−2.98)	0.015 * (1.74)	0.006 (0.66)
Industrial structure (x_3)	0.027 *** (3.28)	−0.010 (−0.50)	0.016 (0.67)
Urbanisation level (x_4)	0.323 *** (5.42)	0.125 (0.93)	0.448 *** (2.87)
Consumption intensity (x_5)	0.129 ** (2.29)	0.535 *** (4.42)	0.664 *** (4.87)
Foreign trade scale (x_6)	−0.109 *** (−5.87)	0.225 *** (6.30)	0.116 *** (2.73)
Observations	620	rho	0.098 **
R-squared	0.445		(2.05)
Number of ids	31	sigma2_e	0.001 ***
Log-likelihood	1135.9894		(17.54)

Notes: z-statistics in parentheses; ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Since spillover effects, spatial heterogeneity in the data, and the incomplete consideration of influencing factors can impact the R-squared value, it is important to note that, despite the R-squared being 0.445 in Table 11, the SDM still demonstrates a relatively high level of fit. This indicates that the explanatory variables adequately reflect the degree of change for the explained variables. The spatial lag coefficient rho is statistically significant at the 5% level, confirming the presence of a significant spatial correlation among provinces. This implies that the coupled development of regional transport logistics and technology is significantly influenced by neighbouring regions. Specifically, a 1% increase in the coupling development level in neighbouring areas leads to a 0.098% increase in the region. Thus, the interlinked growth of transport logistics and technology across China's provinces demonstrates a mutually reinforcing agglomeration effect and spatial spillover effect.

The national economy, industrial structure, urbanisation level, and consumption intensity have a significant positive impact on the coupling coordinated development of the local transport logistics and technology. The urbanisation level is the largest driver, with an increase of 0.323% in the level of coupled development for every 1% increase. While the national economy had the least positive impact. This could be explained by the following reasons. Urbanisation supports the expanding needs of e-commerce and immediate delivery services and intensifies commercial and industrial activities by enhancing transportation infrastructure. This increases the demand for sophisticated logistics services and advanced technology, thereby catalysing the integration of transport logistics and technology. By contrast, while the national economy provides fundamental economic support for transport logistics and technology, it primarily reflects broader macroeconomic conditions. Thus, its impact on transport logistics and technology integration may be overshadowed by more immediate factors.

Population and foreign trade scale have a significant negative impact on the coupling coordinated development of the local transport logistics and technology. In areas with rapidly growing populations, an increased demand places a strain on facilities, reducing logistical efficiency and complicating the adoption of new technologies. In addition, resources might be diverted to expanding basic infrastructure, prioritising capacity expansion over technological innovation. Additionally, foreign trade adds complexity through diverse international regulations and extended supply chains, burdening logistics systems and hindering technological integration. Trade volatility, driven by economic cycles and political changes, introduces unpredictability, complicating planning and investment in technological upgrades. These factors often compel companies to prioritise expanding logistical capabilities to meet immediate demands over advanced transport logistics technologies.

The national economy, population, consumption intensity, and foreign trade scale have a positive impact on the coupling coordinated development of the neighbouring transport logistics and technology. These factors enhance regional economic activities and market demand. This interconnectedness supports upgrades in logistical systems and technological advancements, meeting the demands of evolving consumption patterns and expanded trade. However, the impacts of industrial structure and urbanisation levels on neighbouring cities are not significant. This may be due to their localised influence and the diverse industrial and developmental strategies across different regions.

Given the complexity of the coupled development of transport logistics and technology, effective policy measures should focus on strategic urbanisation, standardised regulations, enhanced regional cooperation, and managed population growth. These measures include strengthening urban infrastructure, such as logistics centres and technology parks, which are crucial for supporting the rising demands of e-commerce and just-in-time delivery services. Governments should streamline regulations to facilitate the integration of technology into logistics systems, thereby enhancing overall efficiency. Simplifying international regulations and shortening supply chains through standardised procedures will strengthen intergovernmental cooperation. Additionally, policies should promote the exchange of best practices and technological innovations across regions, using regional economic activities and market demands to drive the integrated development of transport logistics and technology. Moreover, to address the increasing challenges of urban population growth, policies must focus on managing urban density and enhancing logistics efficiency. For example, optimising the zoning and strategic location of land reduces logistics pressure.

6. Conclusions and Implications

6.1. Conclusions

This paper proposes evaluation index systems for the high-quality development of transport logistics and technology. It utilises the EWM to measure comprehensive development levels and employs a coupling coordination model to assess the coupling coordination degree. Further, kernel density and Moran index analyses are applied to explore the spatial-temporal evolution of the coupling coordination level. Finally, the SDM is used to examine the impact of selected research factors on the coupling coordinated development of transport logistics and technology. The following key findings are established.

Firstly, during the study period from 2003 to 2022, the development level of transport logistics and technology has shown an upward trend across most provinces, with an average annual growth rate of 6.30% for transport logistics and 7.59% for technology. However, significant regional disparities exist, with eastern coastal provinces, like Guangdong, Zhejiang, Shanghai, and Jiangsu, exhibiting higher development levels compared to the western provinces. Notably, the growth in transport logistics generally lags behind

technological advancements, with an increasing number of provinces failing to achieve the desired level of development reciprocity.

Secondly, throughout the study period, most provinces demonstrated a positive upward trend in the coupling coordinated development of transport logistics and technology, with a national average CCD increase of 3.49%. However, the coupling coordination level in most provinces remains at the medium level. From 2003 to 2022, the number of provinces at the medium coupling level rose from 4 to 22, while those at the high coupling level increased to 4. Despite these improvements, regional disparities persist, showing a multi-polar development situation with gradually increasing polarisation and decreasing dispersion. The overall level of coupling coordination decreases from east to west, with provinces at lower development levels struggling to catch up with the leading provinces. Meanwhile, a spatial clustering effect is evident, showing a positive spatial correlation, with HH clusters primarily in the eastern regions and LL clusters in the central–western regions.

Thirdly, the coupling coordinated development of transport logistics and technology in a region is influenced by the neighbouring regions. For every 1% increase in the level of coupled development in neighbouring regions, the region increases by 0.098%. The national economy, industrial structure, urbanisation level, and consumption intensity positively impact the local coupling development, while population and foreign trade have significant negative effects. Urbanisation emerges as the largest driver of coupled development. The national economy, though having a positive influence, exerts the smallest impact compared to other factors.

6.2. Implications

This study examines the coupled development of transport logistics and science and technology, along with their spatial effects and influencing factors, from a spatial–temporal perspective, providing a scientific basis for the formulation of targeted policies and measures. To narrow regional disparities and promote sustainable development, the following recommendations are made to address the challenges in the coordinated development of transport logistics and science and technology.

First, although the eastern coastal provinces lead in the development of transport logistics and technology, the western regions still lag behind. It is recommended that policy guidance and financial support should be strengthened and increased to boost investment in technological innovation and logistics infrastructure in the western regions, thus improving overall regional technological capabilities and logistics efficiency. Additionally, establishing cooperation mechanisms between eastern and western regions can foster regional complementarity in technology and logistics, facilitating the transfer of advanced technologies from the east to the west.

Second, currently, the growth of transport logistics generally lags behind technological advancements. It is suggested that collaborative innovation between the technology and logistics sectors should be enhanced, promoting partnerships between tech companies and logistics firms to jointly develop intelligent and digital logistics technologies. This collaboration would improve overall logistics efficiency. Relevant policies should be developed to accelerate the application of technological innovations in logistics, such as in intelligent transportation and automated warehousing.

Third, research indicates that the coupled development levels of neighbouring regions significantly influence the development of a given region. Therefore, regional cooperation should be enhanced by sharing resources, technologies, and expertise to improve overall coupling coordination. For instance, regional cooperation platforms could be established to promote common development among neighbouring provinces. In particular, such

platforms in central and western regions could help form effective industrial clusters and foster collaborative development across provinces.

Fourth, urbanisation has a significant positive impact on the coupling development of transport logistics and technology. It is recommended that the urbanisation process in small and medium-sized cities should be accelerated, while upgrading local industries by guiding traditional sectors toward high-tech and intelligent solutions. This would promote the deeper integration of technology and logistics. Furthermore, enhancing urbanisation will strengthen the collaborative effect between logistics and technology across regions.

Fifth, the research also reveals that population size and foreign trade have significant negative impacts on coupling coordination development. It is recommended that the population structure should be optimised and there should be more investment in education and training to improve workforce quality. At the same time, foreign trade enterprises should be encouraged to collaborate with local logistics and technology companies to reduce logistics bottlenecks in foreign trade, thereby enhancing the synergy between international logistics and technology.

Author Contributions: Conceptualisation, Q.S.; methodology, Q.S.; software, Q.S.; validation, Q.S., S.J. and H.D.; formal analysis, Q.S. and S.J.; investigation, H.D.; resources, S.J.; data curation, Q.S. and H.D.; writing—original draft preparation, Q.S.; writing—review and editing, Q.S., S.J. and H.D.; visualisation, Q.S.; supervision, S.J.; project administration, S.J.; funding acquisition, S.J. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by Science and Technology Major Project of Guangxi, grant number GuikeAA23062032-1 (Research on Key Technologies for System Engineering and Coordinated Development of the Pinglu Canal) and GuikeAA23062023-3 (Research and Demonstration of Force Prediction and Safe Transportation Reinforcement Technology for Extra-large Transportation Loads in the Pinglu Canal Construction Project).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The data presented in this study are available in the article. The original contributions presented in the study are included in the article, further inquiries can be directed to the corresponding author.

Conflicts of Interest: Author Hanjing Deng was employed by the company Beibu Gulf Port Co., Ltd. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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